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# MD-Bench

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MuCoSim Seminar

# Preliminary

# About

- MD-Bench is a molecular dynamics application which calculates the intersections among particles and how these affect motion
- Simulation system constituents:
  - Number of atoms, each having an initial state and velocity
  - Boundary conditions
- The force of each atom is a sum of its interactions with neighboring atoms

# Force modeling

- In MD-Bench, the Lennard-Jones potential is used for calculating the forces between pairs of particles
  - In our case, particles are electronically neutral atoms
- This function models repulsive as well as attractive interactions

$$V_{LJ}(r) = 4\epsilon\left[\left(\frac{\sigma}{r}\right)^{12} - \left(\frac{\sigma}{r}\right)^{6}\right]$$

# The Lennard-Jones potential I

$$V_{LJ}(r) = 4\epsilon\left[\left(\frac{\sigma}{r}\right)^{12} - \left(\frac{\sigma}{r}\right)^{6}\right]$$

- Where
  - **r** is the distance between two interacting atoms
  - $\epsilon$ is the dispersion energy
  - $\sigma$ is the distance at which the potential **V** is equal to zero

# The Lennard-Jones potential II

- Behavior of the Lennard-Jones potential with increasing distance **r** in **Å** (Ångström)

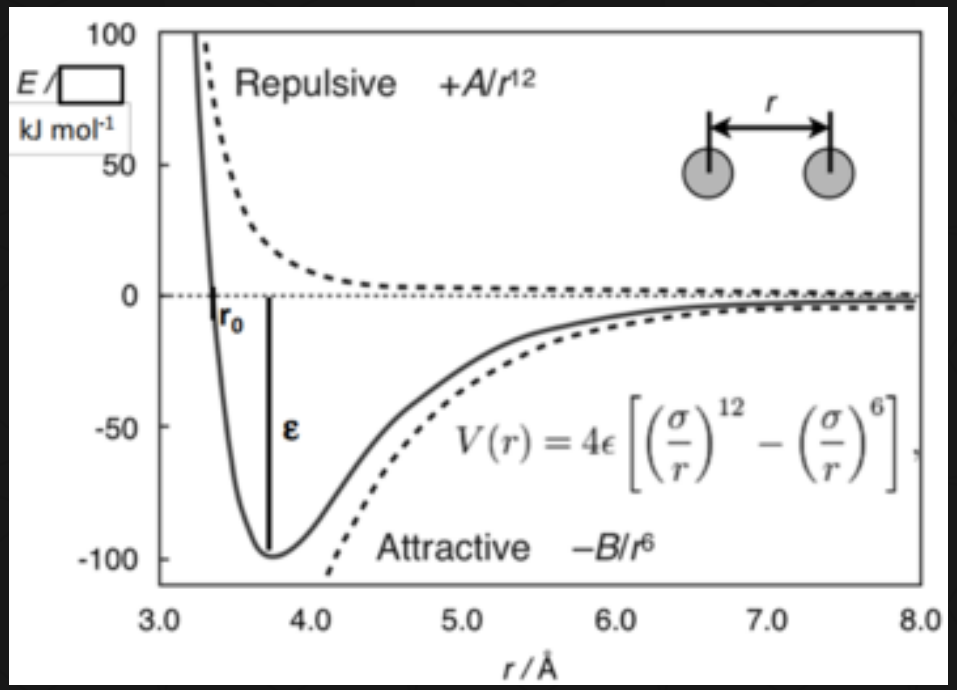



Figure: Lennard Jones Potential

# The Lennard-Jones potential III

- The Lennard-Jones potential is a simplified model but still describes the essential aspects of particle dynamics
- We observe that particles
  - repel each other at close distance
  - attract each other at medium distances and have
  - close to zero interaction at large distances

# Testsystem

# Cluster

- Host/Clustername: tinyGPU.tg086
- Cluster Info
  - https://hpc.fau.de/systems-services/systems-documentation-instructions/clusters/tinygpu-cluster/
- CPU type: 2x Intel Xeon Gold 6226R @2.9 GHz = 32 cores with optional SMT
- GPU type: 8x NVIDIA Geforce RTX3080
- Memory capacity: 384 GB RAM
- Software Environment:
  - Compiler: Nvidia NVCC Build cuda_11.2.r11.2
  - Operating System: Ubuntu 20.04.3 LTS

# MD-Bench

# Force calculation I

- MD-Bench uses the Lennard-Jones potential in order to calculate the forces of atoms [1]
- This calculation is done for pairs of atoms in `src/force.c`

```
for atom in atoms:
    neighbors = neighbor.neighbors[atom]
    force = 0.0
    for neighbor in neighbors:
        radius = calc_radius(...)
        if radius < close_enough:
            force += calc_force(...)
    forces[atom] += force
```

# Force calculation II

- It is time consuming and unnecessary to calculate the force for *every* pair of atoms if they are so far apart that the force is close to 0 anyways

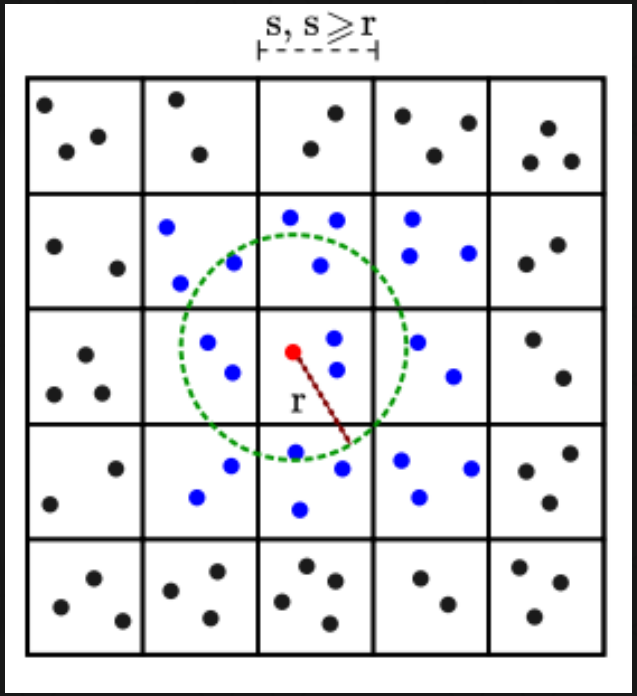



Figure: Neighbor list [1]

# Force calculation III

- src/force.c was refactored to support CUDA kernel functions and distribution among GPUs
- Force-calculation is parallelized on an atom-level
- Each atom (around 100.000) computes the force for every other atom in its neighbor list (around 100)
  - Around $10^7$ force computations per run

# Configuration

- Clone the repository

```
foo@bar:~$ git clone ↙
    ↳ https://github.com/RRZE-HPC/MD-Bench.git
foo@bar:~$ cd MD-Bench
foo@bar:~/MD-Bench$ git checkout mucosim_cuda
```

- Load modules

```
foo@bar:~/MD-Bench$ module load cuda likwid
```

- Change
  - TAG variable in config.mk to TAG ?= NVCC
  - Compiler flags in include_NVCC.mk

# Run Software

- Compile and execute

```
foo@bar:~/MD-Bench$ make
foo@bar:~/MD-Bench$ ./MDBench-NVCC
```

- MD-Bench will output temperature and pressure data as well as performance in million atom updates per second
- Modify number of runs with argument
  -n <nmbr>

# Program Output

- With unmodified parameters

| step | temp | pressure |
|---|---|---|
| 0 | 1.440000e+00 | 1.215639e+00 |
| 100 | 8.200895e−01 | 6.923143e−01 |
| 200 | 7.961495e−01 | 6.721043e−01 |

```
Force field: lj
Data layout for positions: AoS
Using double precision floating point.
```

```
System: 131072 atoms 47265 ghost atoms, ↙
    ↳ Steps: 200
TOTAL 6.83s FORCE 0.73s NEIGH 3.07s REST 3.04s
```

```
Performance: 3.84 million atom updates per ↙
    ↳ second
```

# Scaling runs

# CUDA

- CUDA programming model
  - Basic execution size is a *thread* which is executed by one CUDA core
  - Threads are grouped together as *blocks*, these are executed by streaming multiprocessors (*SM*)
  - Resources on an SM are allocated on a *warp*-size level (32 threads)
  - Warps are dispatched to execution units of GPU

# Scaling threads I

- Default number of atoms: *131072*
- Number of blocks is calculated by (`num_threads` is fixed):

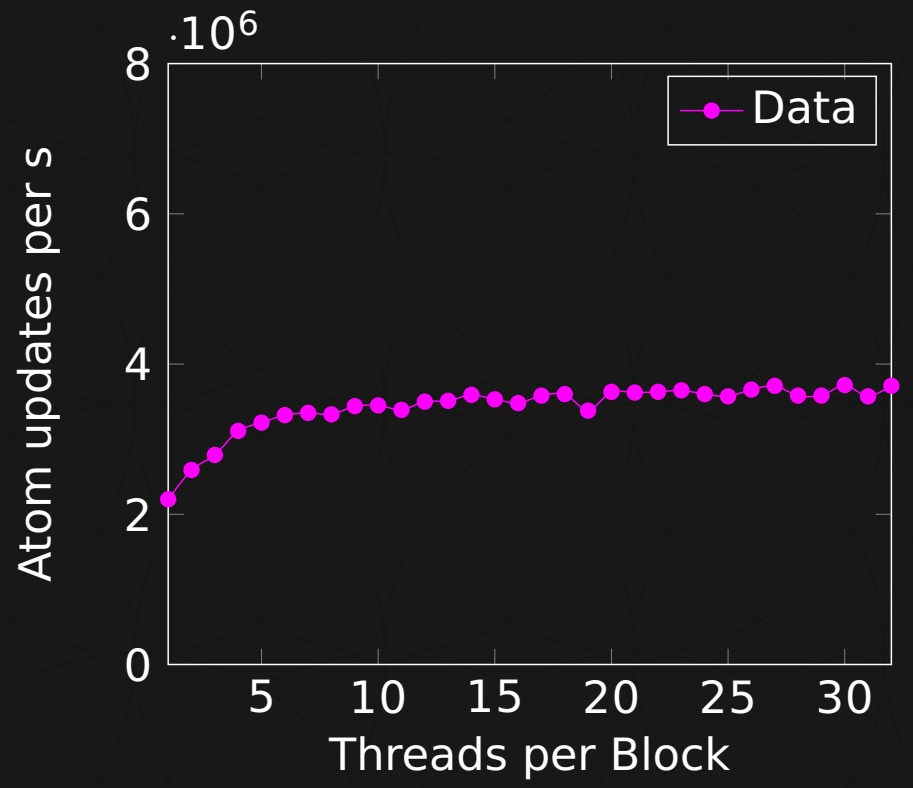
```
blocks = ceil( 131072 / num_threads )
```

- Run on cluster with one GPU:

```
foo@bar:~/MD-Bench$ srun.tinygpu -N 1 -n 1 ↙
    ↳ --gres=gpu:1 --pty /bin/bash -l
foo@bar:~/MD-Bench$ NUM_THREADS=X ↙
    ↳ ./MD-Bench-NVCC -n 50
```

# Scaling threads II

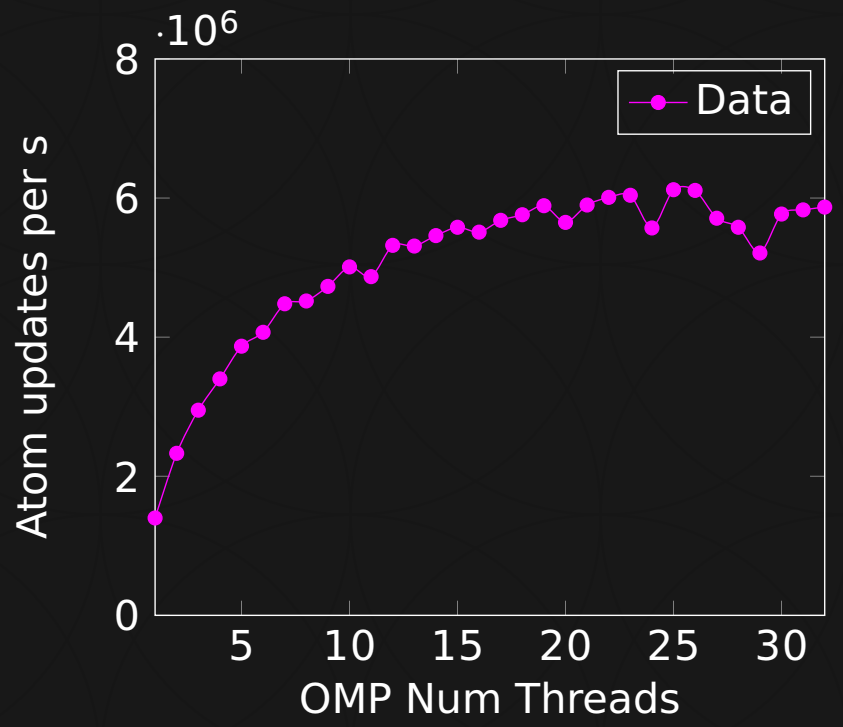
- As resources are allocated on a warp-size level, we expect to have highest performance for 32 threads per block



# Scaling threads III

- Performance improvement stagnates
- But how does the CPU perform?

```
foo@bar:~/MD-Bench$ srun.tinyfat --ntasks=1 ↙
    ↳ --cpus-per-task=32 --pty /bin/bash -l
```



# Measurements

# Memory Bandwidth

- Using the Nvidia *nvprof* profiling tool via

```
foo@bar:~/MD-Bench$ nvprof --print-gpu-trace ↙
    ↳ ./MDBench-NVCC -n 50
```

enables detailed metrics on memory throughput

| Start | Duration | Grid Size | Block Size | Regs | SSMem | DSMem | Size | Throughput | SrcMemType | DstMemType | Device | Context | Stream | Name |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1.00263s | 899.68us | - | - | - | - | - | 4.1199MB | 4.4719GB/s | Pageable | Device | NVIDIA GeForce | 1 | 7 | [CUDA memcpy HtoD] |
| 1.00404s | 87.904us | - | - | - | - | - | 1.0000MB | 11.109GB/s | Pageable | Device | NVIDIA GeForce | 1 | 7 | [CUDA memcpy HtoD] |
| 1.00439s | 87.616us | - | - | - | - | - | 1.0000MB | 11.146GB/s | Pageable | Device | NVIDIA GeForce | 1 | 7 | [CUDA memcpy HtoD] |
| 1.00490s | 87.808us | - | - | - | - | - | 1.0000MB | 11.122GB/s | Pageable | Device | NVIDIA GeForce | 1 | 7 | [CUDA memcpy HtoD] |
| 1.00515s | 60.640us | - | - | - | - | - | 703.13KB | 11.058GB/s | Pageable | Device | NVIDIA GeForce | 1 | 7 | [CUDA memcpy HtoD] |
| 1.00522s | 896ns | - | - | - | - | - | 128B | 136.24MB/s | Pageable | Device | NVIDIA GeForce | 1 | 7 | [CUDA memcpy HtoD] |
| 1.00523s | 864ns | - | - | - | - | - | 128B | 141.29MB/s | Pageable | Device | NVIDIA GeForce | 1 | 7 | [CUDA memcpy HtoD] |
| 1.00524s | 864ns | - | - | - | - | - | 128B | 141.29MB/s | Pageable | Device | NVIDIA GeForce | 1 | 7 | [CUDA memcpy HtoD] |
| 1.00576s | 11.534ms | - | - | - | - | - | 50.000MB | 4.2335GB/s | Pageable | Device | NVIDIA GeForce | 1 | 7 | [CUDA memcpy HtoD] |
| 1.01758s | 45.248us | - | - | - | - | - | 512.00KB | 10.791GB/s | Pageable | Device | NVIDIA GeForce | 1 | 7 | [CUDA memcpy HtoD] |
| 1.01763s | 19.917ms | (65536 1 1) | (2 1 1) | 38 | 0B | 0B | - | - | - | - | NVIDIA GeForce | 1 | 7 | calcforce() [132] |
| 1.03757s | 80.896us | - | - | - | - | - | 1.0000MB | 12.072GB/s | Device | Pageable | NVIDIA GeForce | 1 | 7 | [CUDA memcpy DtoH] |
| 1.03786s | 80.736us | - | - | - | - | - | 1.0000MB | 12.096GB/s | Device | Pageable | NVIDIA GeForce | 1 | 7 | [CUDA memcpy DtoH] |
| 1.03815s | 80.480us | - | - | - | - | - | 1.0000MB | 12.134GB/s | Device | Pageable | NVIDIA GeForce | 1 | 7 | [CUDA memcpy DtoH] |

# Bibliography I

Machado, R. R. L., Schmitt, J., Eibl, S., Eitzinger, J., LeiSSa, R., Hack, S., Pérard-Gayot, A., Membarth, R., and Köstler, H.
tinymd: Mapping molecular dynamics simulations to heterogeneous hardware using partial evaluation.
*Journal of Computational Science 54* (2021), 101425.